

# PRODUCT SALES PERFORMANCE SQL QUERIES

## KPI's:

### A. TOTAL PRODUCTS:

```
SELECT COUNT(Product) AS Total_Product
FROM CustomerPurchaseHistory;
```

| Results Messages |               |
|------------------|---------------|
|                  | Total_Product |
| 1                | 1800          |

### B. TOTAL REVENUE:

```
SELECT SUM(TotalPrice) AS Total_revenue
FROM CustomerPurchaseHistory;
```

| Results Messages |                 |
|------------------|-----------------|
|                  | Total_revenue   |
| 1                | 3266656.8011117 |

### C. TOTAL QUANTITY SOLD:

```
SELECT SUM(Quantity) AS Total_quantity_sold
FROM CustomerPurchaseHistory;
```

| Results Messages |                     |
|------------------|---------------------|
|                  | Total_quantity_sold |
| 1                | 8260                |

### D. AVERAGE ORDER VALUE:

```
SELECT SUM(TotalPrice) / COUNT (DISTINCT Product) AS Average_Order_Value
FROM CustomerPurchaseHistory;
```

| Results Messages |                     |
|------------------|---------------------|
|                  | Average_Order_Value |
| 1                | 466665.257301671    |

## PROBLEM STATEMENT:

### ➤ QUARTERLY PERCENTAGE OF SALES BY PRODUCT CATEGORY:

```
SELECT DATENAME (YEAR, PurchaseDate) AS YearName, DATENAME(QUARTER, PurchaseDate) AS PurchaseMonth,
SUM(Quantity) AS TotalProductSold
FROM CustomerPurchaseHistory
GROUP BY DATENAME(QUARTER, PurchaseDate), DATENAME (YEAR, PurchaseDate);
```

| Results Messages |          |               |                  |
|------------------|----------|---------------|------------------|
|                  | YearName | PurchaseMonth | TotalProductSold |
| 1                | 2023     | 1             | 784              |
| 2                | 2023     | 2             | 828              |
| 3                | 2023     | 3             | 731              |
| 4                | 2023     | 4             | 864              |
| 5                | 2024     | 1             | 717              |
| 6                | 2024     | 2             | 865              |
| 7                | 2024     | 3             | 807              |
| 8                | 2024     | 4             | 949              |
| 9                | 2025     | 1             | 825              |
| 10               | 2025     | 2             | 890              |

### ➤ TOTAL QUANTITY SOLD PER MONTH:

```
SELECT MONTH(PurchaseDate) AS PurchaseMonthNumber, DATENAME(MONTH, PurchaseDate) AS PurchaseMonth,
SUM(Quantity) AS TotalProductSold
FROM CustomerPurchaseHistory
GROUP BY DATENAME(MONTH, PurchaseDate), MONTH(PurchaseDate)
ORDER BY PurchaseMonthNumber;
```

|    | PurchaseMonthNumber | PurchaseMonth | TotalProductSold |
|----|---------------------|---------------|------------------|
| 1  | 1                   | January       | 714              |
| 2  | 2                   | February      | 844              |
| 3  | 3                   | March         | 768              |
| 4  | 4                   | April         | 786              |
| 5  | 5                   | May           | 971              |
| 6  | 6                   | June          | 826              |
| 7  | 7                   | July          | 533              |
| 8  | 8                   | August        | 576              |
| 9  | 9                   | September     | 429              |
| 10 | 10                  | October       | 523              |
| 11 | 11                  | November      | 646              |
| 12 | 12                  | December      | 644              |

### ➤ PRODUCT CATEGORY WISE CUSTOMER COUNT:

```
SELECT ProductCategory, COUNT(CustomerID) AS TotalNumberOfCustomers
FROM CustomerPurchaseHistory
GROUP BY ProductCategory;
```

|   | ProductCategory | TotalNumberOfCustomers |
|---|-----------------|------------------------|
| 1 | Office Supplies | 597                    |
| 2 | Electronics     | 606                    |
| 3 | Furniture       | 597                    |

### ➤ TOTAL REVENUE BY PRODUCT:

```
SELECT Product, CONVERT(DECIMAL(10,2), SUM(TotalPrice)) AS Totalrevenue
FROM CustomerPurchaseHistory
GROUP BY Product
ORDER BY Totalrevenue DESC;
```

|   | Product | Totalrevenue |
|---|---------|--------------|
| 1 | Tablet  | 533091.00    |
| 2 | Laptop  | 481721.40    |
| 3 | Monitor | 468632.80    |
| 4 | Desk    | 454006.70    |
| 5 | Phone   | 444989.72    |
| 6 | Chair   | 443118.71    |
| 7 | Printer | 441096.47    |

### ➤ CUSTOMER PAYMENT TRENDS:

```
SELECT PaymentMethod,
CONVERT(DECIMAL(10,2), ((CountOfPaymentMethod * 100.0/ TotalPayments))) AS PCT
FROM (SELECT PaymentMethod, COUNT(PaymentMethod) AS CountOfPaymentMethod,
(SELECT COUNT(PaymentMethod)
FROM CustomerPurchaseHistory) AS TotalPayments
FROM CustomerPurchaseHistory
GROUP BY PaymentMethod) A
```

|   | PaymentMethod | PCT   |
|---|---------------|-------|
| 1 | Online        | 19.22 |
| 2 | Debit Card    | 22.00 |
| 3 | Gift Card     | 19.61 |
| 4 | Cash          | 19.78 |
| 5 | Credit Card   | 19.39 |

### ➤ Top 3 selling products:

```
SELECT TOP 3 Product, SUM(TotalPrice) AS Revenue
FROM CustomerPurchaseHistory
GROUP BY Product
ORDER BY Revenue DESC;
```

|   | Product | Revenue          |
|---|---------|------------------|
| 1 | Tablet  | 533090.999505997 |
| 2 | Laptop  | 481721.399560928 |
| 3 | Monitor | 468632.800834656 |

### ➤ Bottom 3 selling products:

```
SELECT TOP 3 Product, SUM(TotalPrice) AS Revenue
FROM CustomerPurchaseHistory
GROUP BY Product
ORDER BY Revenue ASC;
```

|   | Product | Revenue          |
|---|---------|------------------|
| 1 | Printer | 441096.470938683 |
| 2 | Chair   | 443118.710647583 |
| 3 | Phone   | 444989.720975876 |

### ➤ Top 3 products by Quantity

```
SELECT DISTINCT TOP 3 Product, SUM(Quantity) AS Quantity
FROM CustomerPurchaseHistory
GROUP BY Product
ORDER BY Quantity DESC;
```

|   | Product | Quantity |
|---|---------|----------|
| 1 | Tablet  | 1316     |
| 2 | Laptop  | 1231     |
| 3 | Monitor | 1180     |

### ➤ Least 3 products by Quantity

```
SELECT DISTINCT TOP 3 Product, SUM(Quantity) AS Quantity
FROM CustomerPurchaseHistory
GROUP BY Product
ORDER BY Quantity ASC;
```

|   | Product | Quantity |
|---|---------|----------|
| 1 | Printer | 1113     |
| 2 | Phone   | 1117     |
| 3 | Chair   | 1147     |

### ➤ Top 3 products by Review

```
SELECT DISTINCT TOP 3 Product, COUNT(*) AS Rating
FROM CustomerPurchaseHistory
WHERE ReviewRating = 5
GROUP BY Product
ORDER BY Rating DESC;
```

|   | Product | Rating |
|---|---------|--------|
| 1 | Tablet  | 69     |
| 2 | Laptop  | 55     |
| 3 | Chair   | 53     |

### ➤ Bottom 3 products by Review

```
SELECT DISTINCT TOP 3 Product, COUNT(*) AS Rating
FROM CustomerPurchaseHistory
WHERE ReviewRating = 1
GROUP BY Product
ORDER BY Rating DESC;
```

|   | Product | Rating |
|---|---------|--------|
| 1 | Laptop  | 57     |
| 2 | Desk    | 57     |
| 3 | Printer | 54     |

## Data cleansing / transformation:

- 1. Quarter column added:** Based on Purchase date, the quarter number is added respectively.
- 2. Correct data type for Purchase date:** Changed the data type of the column purchase date from Whole number to Date format.
- 3. Inserted quarter:** Created a duplicate column for purchase date and changed the data from date to quarter. Renamed column as “Quarter”.
- 4. Product rating:** Created a column that measures the rating numbers.

## Measures:

```
1  DEFINE
2      MEASURE 'Customer product sales'[TotalProducts] = COUNT('Customer product sales'[Product])
3      MEASURE 'Customer product sales'[TotalRevenue] = SUM('Customer product sales'[TotalPrice])
4      MEASURE 'Customer product sales'[TotalQuantitySold] = SUM('Customer product sales'[Quantity])
5      MEASURE 'Customer product sales'[AverageOrderValue] = SUM('Customer product sales'[TotalPrice])/DISTINCTCOUNT
6      ('Customer product sales'[Product])
7  EVALUATE
8      SUMMARIZECOLUMNS(
9          "TotalProducts", [TotalProducts],
10         "TotalRevenue", [TotalRevenue],
11         "TotalQuantitySold", [TotalQuantitySold],
12         "AverageOrderValue", [AverageOrderValue]
13     )
```

### Results

Result 1 of 1 ▾

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|   | [TotalProducts] | [TotalRevenue] | [TotalQuantitySold] | [AverageOrderValue] |
|---|-----------------|----------------|---------------------|---------------------|
| 1 | 1800            | 3266656.8      | 8260                | 466665.26           |